STUDIO UX STANDARD

UX & Menu Visuals Bible

Reusable interaction, accessibility, architecture, and visual-system guidance for Swarmfront and subsequent games.

Normative working standard v0.2 · Swarmfront companion volume · July 15, 2026

UX & Menu Visuals Bible

Status: Normative working standard v0.2
Scope: Menus, dashboards, HUDs, modal flows, blocking in-match and post-match overlays, out-of-match navigation, and reusable UI presentation
Last reviewed: 2026-07-15

1. Purpose and document model

This document is the normative source of truth for how ENTaP games organize, present, and validate player-facing UX.

The documentation family deliberately separates permanent behavior from title-specific and time-sensitive material:

- **This bible** defines required product behavior and portable UX doctrine.
- Swarmfront Game Skin defines Swarmfront's replaceable visual implementation values.
- Swarmfront UI Implementation Status records current adoption, debt, and migration progress.
- New-Title UX Adoption Guide provides the screen brief and Game Skin worksheet.
- Swarmfront UI Sprite Inventory records available and planned artwork.

Normative sections in this bible define required product behavior. Companion documents describe implementations and may change without altering the UX Core.

The goal is not to make every screen or title look identical. The goal is to make every screen feel internally coherent, every title feel deliberately skinned, and every ENTaP product behave according to a shared interaction standard.

2. Requirement language and precedence

2.1 Requirement language

- **MUST** indicates a release requirement. Failure requires correction or an approved exception.
- **SHOULD** indicates the expected implementation. Deviation requires a documented reason.
- **MAY** indicates an optional technique that remains subject to all applicable requirements.

Unqualified declarative rules in the UX Core are treated as MUST requirements unless the text explicitly provides flexibility.

2.2 Order of precedence

When requirements collide, use this order:

1. Player safety, legal requirements, accessibility, and platform requirements
2. UX Core
3. Title-specific Game Skin
4. Screen-specific design decisions

A Game Skin or screen-specific exception MUST NOT weaken navigation safety, accessibility, state clarity, touch geometry, or authoritative-state boundaries.

3. Product promise

Every player-facing screen or blocking overlay MUST answer four questions immediately:

1. **Where am I?**
2. **What can I do here?**
3. **What is the most important action, if any?**
4. **How do I leave or go back?**

If an answer requires experimentation, scrolling without a cue, or force-quitting the application, the screen is not ready.

4. Non-negotiable UX principles

4.1 Clarity before spectacle

Art supports interaction. It MUST NOT compete with labels, obscure state, or determine hit geometry.

- Readability beats ornament.
- A plain working control beats a beautiful ambiguous control.
- Decorative effects MUST remain quieter than interactive state.
- Branding moments MAY be large, but MUST NOT displace the screen's purpose or escape action.

4.2 One screen, one player goal

Every screen has one dominant player goal. A navigational hub MAY offer several peer destinations because its job is to orient and route the player. It MUST NOT simultaneously ask the player to complete unrelated transactions, configure detailed settings, and consume substantial explanatory content.

Examples:

- Jukebox: choose and inspect a map, then play.
- Garage: inspect and equip cosmetics.
- Hive entry menu: create, find, or enter a hive.
- Battle pass: inspect progression and claim rewards.
- Dashboard: orient the player and route them to peer account tools.

4.3 Zero or one dominant action

Each decision step has zero or one visually dominant action. Do not manufacture a primary action on screens whose purpose is browsing, comparison, or orientation.

Other actions are secondary, tertiary, escape, or destructive. Repeated accent color, glow, animation, or oversized art creates fake primaries and destroys hierarchy.

4.4 A visible escape route

Every non-root screen and modal MUST provide an explicit way out.

- The escape control remains visible while content scrolls.
- It is never hidden below a long list.
- It uses text, not an unexplained icon alone.
- System Back/Escape MAY duplicate it, but MUST NOT be the only route.

- Opening a screen MUST NOT trap the player.

4.5 Design for the real device

Screens MUST be reviewed at their actual physical size and expected viewing distance. A desktop editor preview is not release evidence for mobile readability.

- HUD messages, pause flows, match-result overlays, post-match summaries, and their actions use the same physical readability and touch floors as menu UI.
- Font sizes are evaluated after final viewport and content scaling; an authored numeric size alone is not evidence of compliance.
- Developer diagnostics MAY use a compact diagnostic scale in development builds, but MUST NOT carry essential player decisions or ship as the only explanation of player-facing state.

4.6 Progressive disclosure

Show the decision first and explanation second.

- Put the current selection and next action above detail.
- Collapse developer notes and technical metadata.
- Reveal advanced controls only when relevant.
- Avoid presenting an entire feature database as one uninterrupted screen.

4.7 State must be explicit

Selected, available, locked, disabled, loading, empty, pending, success, and error are distinct states. They require distinct copy and presentation.

Color alone MUST NOT communicate state.

5. UX Core versus Game Skin

UX Core: portable	Game Skin: replaceable
Navigation behavior	Logo and key art
Screen hierarchy	Display font
Type roles and physical minimums	Exact typeface pairing and authored values
Touch geometry	Button silhouette
Semantic spacing rhythm	Numeric spacing scale
Button semantics	Palette and glow color
Modal and scrolling behavior	Icons and illustrations
Loading, empty, and error states	Sound and motion personality
Accessibility and QA gates	Branded transitions
State/intent architecture	Game-specific terminology

Future games SHOULD adopt the UX Core and define a new Game Skin. They MUST NOT fork interaction semantics solely to create visual differentiation.

6. Navigation vocabulary

Labels describe outcomes. Use these terms consistently:

Label	Meaning	Behavior
Back	Return to the previous navigational level	Preserve safe, already-applied state
Close	Dismiss a non-sequential overlay	Return to the screen beneath it
Cancel	Abandon an in-progress edit or transaction	Restore the pre-edit value
Done	Finish a flow whose changes apply immediately	Close the flow
Apply	Commit reversible settings without necessarily leaving	Remain or close as stated
Confirm	Approve a consequential decision	Show the consequence in nearby copy
Play / Enter / Buy / Claim	Perform the named domain action	Never replace with generic “OK”

Rules:

- Use **Back** when the player navigated deeper; use **Close** for a non-sequential overlay.
- Back MUST NOT discard unsaved work without warning.
- An icon-only X MUST NOT be the sole escape route on mobile.
- A destructive action SHOULD name the object and consequence: “LEAVE HIVE,” not “CONFIRM.”
- Do not alternate between “Main Menu,” “Home,” and “Dashboard” for the same destination.

6.1 Unsaved-change contract

Every editable flow MUST declare one behavior before implementation:

1. Changes apply immediately and **Done** closes the flow.
2. Changes remain staged until **Apply** or another named commit action.
3. Leaving prompts the player to discard changes or continue editing.

Back, Close, system Back, scene replacement, and app interruption MUST follow the same declared behavior. Closing a flow MUST NOT silently commit or discard changes contrary to that contract.

7. Screen anatomy and navigation structures

The default portrait screen follows this order:

```
┌────────────────────────────────────┐
│ System–safe top / global status    │
│ Screen title                       │
│ Short purpose or current context   │
├────────────────────────────────────┤
│ Primary decision / selected item   │
│                                    │
│ Scrollable supporting content      │
│                                    │
├────────────────────────────────────┤
│ Persistent primary or escape area  │
│ System–safe bottom                 │
└────────────────────────────────────┘
```

Not every screen needs every band. The reading order remains stable:

1. Identity and context
2. Current state or selection
3. Available choices
4. Action, if applicable
5. Escape

7.1 Global navigation

Global navigation is for top-level destinations only. It MUST NOT visually compete with an active task or modal.

When a modal is open:

- Background navigation is inert.
- The background is visibly subordinate.
- Only modal actions receive input or accessibility focus.
- The modal owns a visible Close, Back, or Cancel action.

7.2 Tabs

Tabs switch peer views without changing navigational depth.

- Selected tabs use at least two signals: fill/border plus text or marker.
- Tabs MUST NOT look identical to primary actions.
- Labels remain short; the strip MAY scroll when necessary.
- Text MUST NOT shrink below its minimum merely to fit all tabs.

7.3 Lists and grids

- Prefer one-column lists for descriptive choices.
- Use grids for visually comparable items with short labels.
- Narrow portrait grids SHOULD use no more than three columns for ordinary text buttons.
- Preserve a consistent reading order.
- Make the entire item row actionable, not only its label or icon.

8. Responsive layout standard

8.1 Declared reference canvases

Before menu implementation begins, each title MUST declare exact references for:

- Primary authored canvas or responsive baseline
- Minimum supported physical size
- Maximum supported aspect ratio
- Narrow/tall stress case
- Tablet or large-screen case when supported
- Safe-area configurations
- Text/UI scaling behavior

"Representative device" without a named resolution, scale, safe-area configuration, and physical-size check is not sufficient release evidence. Title-specific values belong in the Game Skin.

8.2 Safe areas

- Interactive controls MUST NOT touch unsafe device edges.
- Persistent bottom actions sit above the system-safe inset.
- Scroll content receives bottom padding equal to the persistent action area.
- Top branding and titles remain below the system-safe top inset.
- Safe-area handling is layout data, never an art-image assumption.

8.3 Reflow before shrink

When space becomes constrained, use this order:

1. Remove low-priority explanatory copy.
2. Stack horizontal panels vertically.
3. Reduce column count.
4. Allow intentional scrolling.
5. Shorten copy without changing meaning.
6. Reduce decoration.
7. Only then consider a modest type reduction, never below the declared physical floor.

8.4 Scrolling

- A scroll region has one clear axis.
- Nested vertical scroll regions are prohibited unless a platform convention requires them.
- Primary and escape actions remain outside long scroll regions.
- Content MUST NOT appear cut off without a scroll affordance.
- Returning to a list SHOULD preserve a useful position.

9. Typography system

Typography is semantic. Code and design refer to roles, not arbitrary pixel sizes.

Role	Purpose	Guidance
Display	Rare branded moments	Logos and event identity; never body copy
Screen title	Names the current destination	One per screen
Panel title	Names a major content block	Short and scannable
Section title	Names a subsection or list	Visually distinct from body
Subtitle	One-line purpose or context	Omit if it adds no information
Body	Decisions, descriptions, values	Primary reading size
Meta	Secondary metadata	Never essential instructions
Button	Action label	Short, active, high contrast
Compact button	Tabs and constrained controls	Not for primary actions

Each title MUST declare numeric values and physical-size floors for these roles. A Game Skin MAY change the numbers and fonts, but MUST preserve the hierarchy and readability contract.

The declared roles apply wherever live player-facing text appears, including HUD status, blocking match overlays, post-match results, and persistent in-game navigation.

- Stylized faces SHOULD be limited to short labels and headings.
- Limited-character fonts MUST NOT be used for dynamic copy.
- Unsupported characters MUST NOT disappear silently.
- Avoid long all-caps paragraphs.
- Avoid outlines and glow on body copy.
- Text baked into art is not a reusable component solution; frame art and live text SHOULD remain separate.

9.1 Copy density

- Screen subtitle: one short sentence.
- Button label: preferably one to three words.
- Panel introduction: one or two lines.
- Technical notes belong in diagnostics, help, or development builds.
- Empty space MUST NOT be filled with prose merely to look occupied.

10. Spacing and touch geometry

Every title MUST define a semantic spacing scale with at least tight, standard, section, panel, and screen-level roles. The rhythm is portable; exact authored units are title-specific.

- Touch targets MUST meet the title's declared physical floor and platform requirements.
- Transparent hit padding MAY enlarge a target whose visible art is smaller.
- Adjacent targets MUST have deliberate separation.
- A target MUST NOT shrink because its icon looks small.
- Important targets SHOULD avoid screen corners and unsafe edges.

11. Button hierarchy, states, and feedback

11.1 Hierarchy

Type	Use
Primary	The next or dominant action in a decision step
Secondary	An alternate safe action
Tertiary	Tabs, filters, and low-emphasis utilities
Destructive	An irreversible or socially consequential action
Escape	Back, Close, or Cancel

11.2 Required states

Every actionable component MUST support applicable states:

- Default
- Hover where pointer exists
- Pressed
- Keyboard/controller/accessibility focus
- Selected
- Disabled
- Loading or pending
- Error

Disabled is not merely lower opacity. It includes a reason when the player could reasonably expect the action to work.

11.3 Feedback and asynchronous actions

- Press feedback begins immediately.
- Success or failure appears near the initiating action.
- Toasts supplement state changes; they do not replace persistent state.

Once an asynchronous action is accepted:

- The initiating control prevents duplicate submission.
- The UI preserves enough context to display the canonical result.
- Leaving the screen MUST NOT manufacture success, failure, or cancellation.
- Re-entering the screen renders authoritative current state.
- A presentation timeout MUST NOT imply transaction failure unless the authoritative operation reports failure.

12. Color and surface system

- Body text targets at least 4.5:1 contrast against its actual background.
- Large text and essential non-text controls target at least 3:1.
- Decorative glow does not count toward contrast.

- Color is paired with text, shape, icon, or position to indicate state.
- Interactive surfaces are distinguishable from decorative frames.
- Background texture remains below the contrast needs of foreground content.
- Dense text sits on a stable surface.
- Important labels MUST NOT sit directly over high-frequency art.

Exact palette, material language, and surface recipes belong in the Game Skin.

13. Iconography and imagery

- Icons reinforce labels; they MUST NOT replace unfamiliar actions.
- Use one icon metaphor per action across a title.
- Maintain a consistent stroke, perspective, material, and lighting family.
- Selected and locked overlays remain separate state layers, not baked variants of every asset.
- Key art uses aspect-preserving containers and MUST NOT define interaction bounds.
- Crop intentionally; never stretch logos or character art.
- Essential information MUST NOT exist only inside an image.

14. Modal, drawer, and overlay contracts

14.1 Modal

Use a modal for a focused decision that temporarily blocks the parent screen.

- Clear title and purpose
- Zero or one primary action
- One explicit escape action
- Dimmed, inert background
- Bounded content with intentional scrolling when needed
- Persistent action area outside long content

14.2 Drawer

Use a drawer for peer navigation or a workspace that benefits from retaining background context.

- Drawer position remains consistent.
- Opening or closing does not change gameplay state.
- The drawer owns a visible handle or Close/Back action.
- Background input behavior is explicit.

14.3 Full-screen destination

Use a full-screen destination for deep browsing, progression, commerce, or tasks with multiple subviews. It receives its own screen title and Back route.

15. Content-state contract

Every data-driven surface defines these states before visual polish:

State	Required presentation
Loading	Stable skeleton/progress and plain-language status
Empty	What is empty, why it matters, and the next useful action
Error	What failed, what remains safe, and retry/back options
Offline	Which features remain available and which require connection
Locked	Requirement and, when appropriate, route to unlock
Pending	Prevent duplicate input and show that work continues
Success	Updated state plus a clear next action when needed

Raw identifiers, debug labels, and placeholder “Unknown” copy MUST NOT appear in a final player-facing state when a graceful fallback is possible.

16. Motion and sound

Motion explains change and reinforces hierarchy.

- Small state transitions SHOULD be brief.
- Frequent navigation MUST NOT use long blocking reveals.
- Motion SHOULD animate position, opacity, or emphasis with a clear purpose.
- Decorative elements MUST NOT all animate simultaneously.
- Reduced-motion behavior is required.
- UI timing is presentation-only and MUST NOT mutate or infer gameplay state.

Sound follows the same hierarchy: restrained navigation, clear confirmation, distinct warning, and no repeated noise during scrolling.

Exact duration, easing, and sound values belong in the Game Skin.

17. Copy, localization, and dynamic content

Interface copy is direct, specific, and player-facing.

- Start buttons with verbs: Play, Browse, Claim, Equip, Leave.
- Prefer concrete nouns: Hive, Map, Loadout, Reward.
- Remove implementation language such as “profile-backed,” “scaffold,” and raw state names.
- Explain restrictions next to the restricted action.
- Use sentence case for prose. Branded all-caps MAY be used for short navigation labels.
- Numbers use consistent separators, units, currency formatting, and time formats.
- Novelty copy MUST NOT obscure a purchase or destructive decision.

Player-facing components MUST tolerate:

- At least 30–50% text expansion
- Long player, group, map, and event names
- Plural and variable-number substitution

- Missing-glyph fallback
- Right-to-left layout where the title commits to supporting it

Truncation MUST be intentional and MUST NOT hide the distinction between consequential actions.

18. Input and focus contract

- Every actionable flow MUST work with each supported input mode.
- Opening a modal moves focus into it.
- Focus MUST NOT escape to inert background controls.
- Closing a modal restores focus to the invoking control or nearest valid replacement.
- System Back/Escape resolves only the topmost active layer.
- One physical input event MUST trigger at most one semantic action.
- Touch, mouse, controller, keyboard, and accessibility paths MUST produce equivalent outcomes.
- Input-mode changes SHOULD preserve the player's current context and selection.

19. Accessibility baseline

Every release candidate MUST support:

- Readable text at target physical size
- Required contrast
- Declared touch target floor
- Visible focus state
- Logical focus and reading order
- Color-independent state communication
- Accessibility labels where supported
- Reduced motion
- No essential information that exists only in an image
- Dynamic text tolerance or a documented platform constraint

Accessibility is part of component definition, not a final polish pass.

20. Technical architecture contract

20.1 Authoritative state

- Simulation or operations state is the source of truth.
- UI reads authoritative state and renders it.
- UI emits intents or commands.
- UI, input, and render layers MUST NOT mutate gameplay state directly.
- Visual state MUST NOT be used to infer gameplay state.
- Opening, closing, or animating a menu has no hidden gameplay side effect.
- Presentation caches are disposable and never become a second authority.

Reusable components SHOULD receive semantic inputs such as `primary`, `selected`, `locked`, or `loading`, rather than reconstructing styles from raw values on every screen.

20.2 Permissible UI-local state

UI MAY own disposable view-local state such as:

- Focus and hover
- Scroll position
- Expanded or collapsed sections
- Temporary animation progress
- An explicitly uncommitted edit buffer

View-local state MUST NOT become evidence that a gameplay, economy, identity, purchase, or progression operation succeeded.

20.3 Analytics boundary

Analytics observes UI behavior. It MUST NOT determine whether an action succeeds, block navigation, delay feedback, or become authoritative product state. Analytics failure MUST be invisible to the player.

21. Menu performance contract

- Animation MUST NOT delay input availability unless the transition intentionally blocks interaction.
- Decorative animation MUST NOT cause sustained node, material, texture, or memory growth.
- Repeated opening and closing MUST return to a stable resource baseline.
- Loading remote data MUST NOT block Back, Close, or safe local navigation.
- Performance validation MUST use representative low-end target hardware.
- A title SHOULD define measurable frame-time, memory, and load-time budgets before release hardening.

22. Review and iteration loop

Use this loop for Ralph-style visual iteration or any human/AI design pass:

1. Capture the current screen at every declared reference size.
2. Write the screen's player goal and primary action, if any, in one sentence.
3. Mark failures by severity:
    - **P0**: trapped player, inaccessible action, purchase/destructive ambiguity
    - **P1**: clipped essential copy, unreadable decision, broken state feedback
    - **P2**: inconsistent hierarchy, density, spacing, or terminology
    - **P3**: cosmetic polish and atmosphere
4. Fix navigation and comprehension before ornament.
5. Change a small, named set of variables per pass.
6. Capture the same states and sizes again.
7. Run interaction, focus, and layout tests.
8. Record accepted UX decisions here and visual decisions in the Game Skin.

Do not judge a pass solely from a static beauty shot. Verify normal, selected, disabled, loading, pending, empty, error, localized, focused, and scrolled states.

23. Release acceptance checklist

A menu is release-ready only when all applicable items pass.

Purpose and hierarchy

- [ ] The destination is named.
- [ ] The player goal is understandable within a few seconds.
- [ ] There is zero or one dominant action per decision step.
- [ ] Essential information is visually stronger than decoration.

Navigation and editing

- [ ] A visible Back, Close, or Cancel route exists.
- [ ] The escape route remains reachable while content scrolls or loads.
- [ ] System Back/Escape resolves the topmost active layer.
- [ ] Background controls cannot activate through a modal.
- [ ] Editable flows declare and honor their unsaved-change behavior.

Layout and readability

- [ ] Every exact title reference size and safe-area case has been reviewed.
- [ ] Essential text and controls do not clip.
- [ ] Type and touch geometry meet declared physical floors.
- [ ] Scrolling is obvious and intentional.
- [ ] Localization expansion and long dynamic names have been tested.

Input and accessibility

- [ ] Supported input modes produce equivalent outcomes.
- [ ] Focus enters, remains within, and exits modals correctly.
- [ ] One physical event triggers no more than one semantic action.
- [ ] Selected and disabled states use more than color alone.
- [ ] Reduced-motion behavior works.

State and feedback

- [ ] Applicable loading, empty, error, offline, locked, pending, and success states exist.
- [ ] Asynchronous actions prevent duplicates and render canonical results.
- [ ] Failed actions explain recovery.
- [ ] Presentation timeout does not manufacture authoritative failure.

Architecture, performance, and verification

- [ ] UI reads authoritative state and emits intents only.
- [ ] No gameplay rules live in menu or render code.
- [ ] Analytics failure cannot affect player outcomes or navigation.
- [ ] Repeated open/close cycles return to a stable resource baseline.

- [ ] Layout, interaction, focus, and screenshot tests pass.
- [ ] Player-facing copy contains no debug language or raw identifiers.

24. Governance

- This bible owns interaction and presentation behavior for menus.
- Game-specific canon owns terminology and fiction.
- Simulation specifications own gameplay rules.
- Game Skins own title-specific visual values but MUST conform to this bible.
- Asset inventories describe artwork and do not override UX behavior.
- Exceptions require a player benefit, affected screens, violated requirement, test plan, approver, and review date.
- A repeated exception SHOULD become a component or revised core rule, not permanent screen-local handling.

When this document and an existing screen disagree, treat the screen as migration work unless a newer approved decision explicitly supersedes the bible.

Swarmfront Game Skin

Status: Non-normative title implementation v0.1
Governed by: UX & Menu Visuals Bible
Last reviewed: 2026-07-15

This document defines Swarmfront's replaceable visual implementation. It may evolve without changing UX Core behavior.

Identity

Swarmfront's interface language is forged dark metal, restrained honey-gold energy, and hex-derived geometry.

The emotional target is:

- Tactical
- Industrial
- Charged

The interface should not glow everywhere. Gold is a scarce hierarchy signal.

Reference canvases

Class	Reference	Purpose
Authored portrait	1080 × 1920	Primary Swarmfront viewport
Narrow/tall stress	944 × 2048	Clipping and hidden-exit validation
Small portrait	720 × 1280	Density and minimum-size validation
Tablet/large	Must be declared before tablet release	Line length and dead-space validation

Godot currently uses `viewport` stretch with `keep_width` aspect behavior.

Typography

Primary interface face: Iceland
Limited display treatments: specialized atlas fonts only where their character set is known and static

Current shared tokens live in `res://scripts/ui/ui_typography.gd`.

Menu and dashboard portrait surfaces use the shared 2× scale. Player-facing pre-match, in-game, and post-match surfaces use a 2.5× scale—25% larger—because attention is split between UI and the playfield and the device is commonly held farther away.

Token	Base size	Menu/dashboard portrait 2×	Pre/in/post-match portrait 2.5×
`screen_title`	28	56	70
`panel_title`	24	48	60
`section_title`	18	36	45

Token	Base size	Menu/dashboard portrait 2×	Pre/in/post-match portrait 2.5×
`panel_subtitle`	17	34	43
`body`	16	32	40
`meta`	15	30	38
`button`	17	34	43
`compact_button`	15	30	38

Atlas fonts must not receive dynamic player-facing copy. Frame art and live text should remain separate.

Spacing and touch

Swarmfront uses a 4-unit authored-canvas rhythm:

Token	Value	Typical use
`space_1`	4	Tight internal relationships
`space_2`	8	Button groups and compact rows
`space_3`	12	Standard sibling separation
`space_4`	16	Control padding
`space_6`	24	Panel padding and section gaps
`space_8`	32	Major section separation
`space_12`	48	Screen-level separation

Current mobile control floor: 64 authored units high
Standard menu actions: 64–72
Persistent or high-consequence actions: 88–112

These are Swarmfront authored-canvas values, not automatic numeric requirements for another title.

Working palette

Role	Approximate value	Use
Deep background	`#0A0A0D`	Page foundation
Panel background	`#14171F` at high opacity	Readable content surfaces
Elevated popup	`#0F121A` at near-opaque	Modal/dropdown separation
Steel edge	`#595C70`	Secondary boundaries
Honey gold	`#F7BA30`	Primary emphasis and currency
Warm text	`#FFF0B8`	Primary action labels

Role	Approximate value	Use
Cool body text	`#EBEDF3`	General reading
Muted text	`#AEB3C0`	Nonessential metadata only
Destructive red	Muted dark red plus light red edge	Consequential actions

Control treatments

Type	Swarmfront treatment
Primary	Warm dark fill, gold edge, brightest readable label
Secondary	Cool charcoal fill, steel edge, light label
Tertiary	Quiet dark surface; compact but readable
Destructive	Restrained red edge/fill plus explicit consequence copy
Escape	Visually secondary, persistent, and easy to locate

Interactive state must remain stronger than seams, particles, and decorative glow.

Backgrounds and surfaces

- Dense text sits on a stable near-opaque panel.
- Hex seams, particles, and animated energy are atmosphere, not navigation.
- Important copy does not sit directly over logos or high-frequency texture.
- Store, Hive, Dash, and Popup contexts currently have distinct hex background presets.

Imagery and assets

- Canonical logo source: `res://assets/branding/swarmfront_logo_1024.png`
- Sprite inventory: SF UI Sprites V1
- Background presets: `res://ui/backgrounds/HexBgPresets.gd`
- Shared typography: `res://scripts/ui/ui_typography.gd`

Selected and locked state layers should remain independent from base art. Logos and character renders preserve aspect ratio and never define hit geometry.

Motion and sound direction

Motion should feel mechanical and energized without becoming ceremonial during frequent navigation.

- State response is immediate.
- Panel transitions are short and purposeful.
- Persistent ambient movement remains quieter than selection or confirmation.
- Reduced-motion mode uses immediate updates or restrained crossfades.

- Navigation, confirmation, and warning sounds remain distinct.

Exact timing and sound values remain open for device tuning.

Skin prohibitions

Swarmfront's skin must never:

- Hide or displace an escape route
- Use gold on every action
- Bake dynamic or localized labels into art
- Make decoration look more actionable than controls
- Use unsupported display-font characters for dynamic copy
- Override UX Core semantics to preserve a visual composition

Swarmfront UI Implementation Status

Status: Time-sensitive implementation record
Governed by: UX & Menu Visuals Bible
Skin: Swarmfront Game Skin
Snapshot date: 2026-07-15

This record may become stale without changing the validity of the normative UX standard.

Established

- Canonical 1080 × 1920 portrait viewport and `keep_width` behavior
- Canonical Swarmfront logo source
- Shared semantic typography tokens in `UITypography`
- 64-unit baseline mobile touch target
- Larger Dashboard tabs and readable Garage category grid
- Portrait stacking for Buffs and Achievements dashboard panels
- Enlarged primary Hive pull-down with explicit Back action
- Persistent bottom “BACK TO MAIN MENU” action in Jukebox
- Identity-first pre-match presentation with redundant map/mode facts removed; countdown and actionable setup prompts remain
- Content-hugging post-match modal with scrollable details and an always-visible responsive action footer
- Portrait smoke coverage for Dashboard, Hive, Jukebox, and dense post-match layout/exit behavior
- Dark panel, steel edge, and honey-gold emphasis direction
- Hex background presets for Store, Hive, Dash, and Popup contexts

Partial adoption

- Typography tokens exist but are not used by every menu.
- Button styling is split among raw Godot buttons, `UIButton`, `HexButton`, sprite-backed controls, and local helpers.
- Color roles are recognizable, but values remain duplicated across scripts.
- Several dialogs meet touch sizing, while older utilities still use 30–52-unit controls.
- Main flows are portrait-first, but tablet/large-screen references are not declared.
- Some art buttons contain baked text, limiting responsive layout and localization.
- Modal behavior is implemented screen by screen rather than through one shared template.

Known migration debt

- Complete type-token adoption across menus and dialogs
- Consolidate a semantic theme/component layer
- Normalize Back, Close, Cancel, Apply, and Confirm usage
- Guarantee persistent escape actions on every overlay
- Remove clipped copy, raw identifiers, and development language
- Add unified focus/controller behavior and restoration
- Add safe-area tokens instead of screen-specific margins

- Standardize loading, empty, offline, locked, pending, and error components
- Separate localized live labels from artwork
- Add localization expansion and missing-glyph testing
- Complete reduced-motion and accessibility audits
- Add menu open/close resource-baseline tests
- Automate screenshot matrices across declared reference sizes

Component migration order

1. **Type roles** — one shared token source
2. **Action button** — primary, secondary, tertiary, destructive, escape
3. **Panel surface** — standard, elevated, modal, selected
4. **Screen frame** — title, context, content, persistent action/safe area
5. **Tabs and segmented control** — selected/focus/disabled states
6. **List/grid item** — selected, locked, pending, empty
7. **Modal template** — title, focus trap, scroll body, primary, escape
8. **Status components** — loading, empty, error, offline
9. **Toast/banner** — supplemental event feedback
10. **Focus/accessibility layer** — input-mode-independent behavior

Component code owns behavior and semantic styling. Game Skin resources supply palette, typography, imagery, sound, and motion values.

Current exclusions

- Map topology or map visual redesign
- In-match lane, hive, unit, or capture-rule presentation
- Economy or progression rule changes
- Gameplay-state architecture changes

Updating this record

Update the snapshot date whenever an item changes category. Do not modify the normative bible merely to make this status record match an implementation shortcut.

New-Title UX Adoption Guide

Status: Maintained adoption guide
Governed by: UX & Menu Visuals Bible

Do not begin a new title by copying Swarmfront's finished screens. Adopt the UX Core and create a new Game Skin.

Adoption sequence

1. Adopt the semantic type roles, spacing rhythm, navigation vocabulary, state model, and release gates.
2. Declare exact target devices, reference canvases, viewport strategy, safe areas, and accessibility targets.
3. Complete the Game Skin worksheet.
4. Build semantic components with neutral placeholder styling.
5. Prove one representative screen at every declared reference size.
6. Apply the Game Skin to components, not individual screens.
7. Validate loading, empty, error, locked, disabled, selected, localized, focused, and scrolled states.
8. Add screenshot, focus, resource-baseline, and interaction tests before multiplying screens.

Game Skin worksheet

```
Title:
Audience and platform:
Emotional promise (three words):
World/material metaphor:

Primary typeface:
Secondary/readability typeface:
Display-only typeface or treatment:

Page background:
Panel surface:
Elevated/modal surface:
Primary accent:
Secondary accent:
Body text:
Muted text:
Success:
Warning:
Destructive:

Primary control silhouette:
Secondary control silhouette:
Panel geometry:
Icon style:
Illustration/render style:

Default motion character:
Reduced-motion behavior:
Navigation sound:
Confirm sound:
Warning sound:

Primary authored canvas/responsive baseline:
Minimum supported physical size:
Maximum supported aspect ratio:
Narrow/tall stress reference:
Tablet/large-screen reference:
Safe-area configurations:
Text/UI scaling behavior:
Minimum physical touch target:
Contrast target:

Localization expansion target:
Right-to-left commitment:
Low-end performance target:
```

```
What this skin must never do:
Three reference screens:
```

Screen design brief

Every new menu starts with this brief before polished art:

```
Screen:
Player goal:
Entry points:
Exit destination:
Primary action, if any:
Secondary actions:
Unsaved-change behavior:
Information required to decide:
Loading state:
Empty state:
Error/offline state:
Locked state:
Pending/async behavior:
Scrollable region:
Persistent controls:
Initial focus:
Focus restoration target:
Supported input modes:
State read from:
Intent(s) emitted:
Reference sizes and safe areas tested:
Localization stress cases:
Accessibility considerations:
Performance considerations:
Analytics/telemetry events:
```

If the player goal or exit destination is unclear, the screen is not ready for visual design. A browsing or orientation screen may correctly have no primary action.

Portability test

The system is portable when a new palette, font family, silhouette, icon set, and motion profile can be applied without rewriting navigation behavior, screen state, or interaction tests.

If a visual change requires changing the meaning of Back, selected, disabled, loading, pending, or primary, the Game Skin has crossed into UX Core and needs explicit review.

Adoption completion

Adoption is complete when:

- The title has a reviewed Game Skin.
- Exact reference canvases and physical checks are recorded.
- Core components implement supported input, focus, accessibility, and content states.
- One representative flow passes the normative release checklist.
- Analytics can fail without affecting the experience.
- Repeated menu use returns to a stable resource baseline.

SF UI SPRITES — INVENTORY (MENU + DASH FAMILY)

> This is an asset inventory. Interaction, hierarchy, navigation, typography, and component behavior are governed by UX & Menu Visuals Bible.

Status legend: DONE / NEXT / LATER

A) DONE — ALREADY CREATED (LOCKED)

PRIMARY MENU / NAV

- [ ] DASH button (left variant) — Status: NEXT — Icon concept: half-hex tab (left cut) — Path: (not created)
- [ ] DASH button (right variant) — Status: NEXT — Icon concept: half-hex tab (right cut) — Path: (not created)
- [ ] PLAY button — Status: NEXT — Icon concept: bold hex/plate, play triangle — Path: (not created)
- [ ] STORE button — Status: NEXT — Icon concept: hex/plate, tag/box glyph — Path: (not created)
- [ ] TIME PUZZLE button — Status: NEXT — Icon concept: hex/plate, hourglass/puzzle glyph — Path: (not created)
- [ ] HIVE button — Status: NEXT — Icon concept: hex/plate, hive glyph — Path: (not created)
- [ ] SETTINGS button (single, isolated file) — Status: NEXT — Icon concept: gear glyph — Path: (not created)

SUBMENU / UTILITY

- [ ] STATS button — Status: NEXT — Icon concept: bar chart glyph — Path: (not created)
- [ ] ANALYTICS button — Status: NEXT — Icon concept: waveform/graph glyph — Path: (not created)
- [ ] REPLAY button — Status: NEXT — Icon concept: replay arrow glyph — Path: (not created)
- [ ] BUFFS button (shield icon variant) — Status: NEXT — Icon concept: shield glyph — Path: (not created)
- [ ] BUFFS button (flex arm icon variant) — Status: NEXT — Icon concept: flex arm glyph — Path: (not created)
- [ ] RULES button — Status: NEXT — Icon concept: document glyph — Path: (not created)
- [ ] BACK button — Status: NEXT — Icon concept: left arrow glyph — Path: (not created)

B) NEXT — MUST-HAVE (BLOCKERS IF MISSING)

1. CONFIRM / APPLY (choose ONE label and reuse everywhere)

- [ ] Label: CONFIRM (or APPLY or ACCEPT) — Icon: checkmark/stamped seal — Constraint: metallic/hex family, yellow text

2. CANCEL / CLOSE (distinct from BACK)

- [ ] Label: CANCEL (or CLOSE) — Icon: X or hex-collapse mark — Constraint: NOT an arrow

3. INFO / HELP (SMALL ICON BUTTON)

- [ ] Label: icon-only — Icon: ⓘ or ? — Constraint: small glyph-style sprite

4. LOCKED / UNLOCKED OVERLAY ICONS (NOT BUTTONS)

- [ ] Label: LOCK — Icon: closed lock — Constraint: neutral/cold

- [ ] Label: UNLOCK — Icon: open lock — Constraint: neutral/cold

C) LATER — SHOULD-HAVE (V1.1, EASY TO ADD)

- [ ] NEXT / FORWARD — Icon: arrow right — Constraint: multi-page flows
- [ ] EXPAND / COLLAPSE — Icon: chevrons up/down or hex fold — Constraint: low-profile
- [ ] FILTER / SORT — Icon: funnel or sliders — Constraint: small glyph
- [ ] SEARCH — Icon: magnifier — Constraint: small glyph

D) LATER — FUTURE / MODE-SPECIFIC (DO NOT BUILD NOW)

- [ ] INVITE / SHARE — Icon: share glyph — Constraint: optional
- [ ] WATCH / SPECTATE — Icon: eye/play glyph — Constraint: distinct from Replay
- [ ] REPORT / FEEDBACK — Icon: flag/exclaim glyph — Constraint: tiny, buried

E) OUT OF SCOPE

- In-match sprite family is out of scope for this list (hives, lanes, buff indicators, timers, alerts).